CHINS 111

OVEN INSTALLATION

READ INSTRUCTIONS BEFORE DOING ANYTHING!

INSPECTION

Check for damage during shipment BEFORE unloading the machine from the truck. If there is any visible damage, report it immediately to Casa Herrera and the freight company. Check the shipping list with the number of crates, packages and parts to be sure that everything has arrived. If there are any shortages, report them to Casa Herrera immediately.

SET THE MACHINE IN PLACE

Unless locally-applied codes specify otherwise, recommended minimum clearance is 36 inches on all sides except for the front, which should have 48 inches clearance. Adequate room must be provided for servicing and proper operation. Install the pipe legs or leveling screws as provided. It is very important that the oven be level in all directions, to assure even baking belt tracking. The oven area must be kept free and clear from combustibles.

DRAFT HOOD and AIR SUPPLY

Install a UL-approved draft hood that satisfies with local code requirements. Or, use a proper-sized draft diverter, installed 6 to 8 feet from the top of the oven exhaust. DO NOT USE DAMPERS! See drawing C79026 in this manual. DO NOT INTERCONNECT WITH OTHER EQUIPMENT!

THIS UNIT IS NOT FOR CONNECTION TO A TYPE-B VENT

CAUTION! Do not obstruct the flow or supply of fresh air (sometimes called “makeup air”) to the oven area from outside, by tightly shutting up the room or otherwise preventing fresh air from freely flowing in to replace air used by the oven. Also, all covers and baffles provided by CASA HERRERA as integral parts of this oven must be in place and in good condition. Exhaust air must be properly vented to the outside according to code requirements. Failure to observe this direction could result in concentrations of noxious and dangerous products of combustion.

GAS LINES

The installation must conform with local codes, or in the absence of such, the National Fuel Gas Code, ANSI Z223.1:

The oven and its individual shutoff valve must be disconnected from the gas supply piping systems during any pressure testing of that system at test pressures in excess of 1/2 psig (3.45 KPa)
The oven must be isolated from the gas supply piping system by closing its individual manual shutoff valve during any pressure testing of the gas supply piping system at test pressures equal to or less than 1/2 psig (3.45 Kpa).

This oven must be installed with a gas line the same pipe size as the oven's Gas Safety Shutoff Valve, and a pressure regulator capable of being adjusted to an outlet pressure of 14 inches water column. The gas line should also have a "drip leg" to avoid pipe scale and foreign materials from getting into the seat of the gas solenoid valve. It is also recommended that the gas vent line be piped to an outside location. Gas pressure needed is between 11 to 14 inches W/C under flow conditions, not static (no gas being used). Check the oven nameplate.

ELECTRICAL REQUIREMENTS AND INSTALLATION

Electrical diagrams for the oven/sheeter are located in the **"OVEN DRAWINGS**" section of this manual. Also, a copy was placed in the electrical system enclosure prior to shipment.

Verify that **VOLTAGE, FREQUENCY** and **AMPERAGE** capacity of your electrical power supply is fully compatible with the information on the machine nameplate. Connect the electrical supply according to code requirements. **THIS MACHINE MUST BE SECURELY EARTH-GROUNDED**, according to local codes, or in the absence of such, the National Electric code, ANSI/NFPA 70. If the product conveyor belts run backward, adjust the supply wiring. **DO NOT CHANGE THE UNIT'S INTERNAL WIRING!**

GEAR OIL LEVEL

The oil level in the gear boxes is normally checked and serviced before shipping. Always check before start-up, **JUST TO MAKE SURE!**

CHINS 112

OVEN OPERATION & SAFETY CHECK

PRE-MIX OVEN OPERATION

NOTE: THESE INSTRUCTIONS TELL YOU WHAT TO DO AND WHAT SHOULD HAPPEN.

IF SOMETHING DOESN'T HAPPEN AS DESCRIBED, STOP THERE! TURN THE MACHINE "OFF". THEN CALL YOUR SUPERVISOR OR MAINTENANCE.

PRE-OPERATION SAFETY CHECK

Check that all guards are in place and secured. Make sure that oven side panels and doors are in place and good condition and closed.

Look in both ends of the oven to see if anything has been left on any of the belts. Also, make sure that no one is working on or around the machine you are about to start.

STARTING THE OVEN

1. Turn the ELECTRICAL SAFETY FEEDER DISCONNECT TO "ON".
2. Turn MAIN POWER switch "ON".
3. Pull all emergency stops to "On" position or start oven drive from OIT (Operators Interface Terminal). All the bake belts should be moving.
4. Set the TEMPERATURE CONTROL to the desired temperature.
5. Pull the (combustion) BLOWER switch "ON".
6. Pull the HEAT START switch "ON". This starts a timed purge cycle. When completed, the spark ignitor is energized and a 5-second trial for ignition begins. If the pilot flame is proven, the main burners will ignite within 10 seconds. The oven will rise to set point temperature.

NOTE: If the burners do not ignite or go out after lighting, push the RESET button for the FLAME SAFETY CONTROL and do again as directed in "6". If the burners do not stay lit after three tries, notify your supervisor or maintenance department before trying again.

SHUTTING OVEN OFF

1. Allow the remaining product to finish its trip through the oven.

2. Push the HEAT START switch “OFF”.

3. Allow the oven belts to run until the oven has cooled to about 250 °F

4. Push (combustion) BLOWER switch “OFF”.

5. Push the BELT-DRIVE switch to “OFF”.

6. Turn the ELECTRICAL SAFETY DISCONNECT switch to “OFF”.

DWELL TIME (BAKE BELT) ADJUSTMENT

Electrical: AC Variable-Frequency Controller feeds bake belt motor. Adjust speed pot at operator panel or digital OIT speed control.

Oven Operation

Turn the oven belts on at the touchscreen (Figure 1.1). Check to see if it is clear of people and we also recommend a verbal announcement to confirm ‘All Clear’.

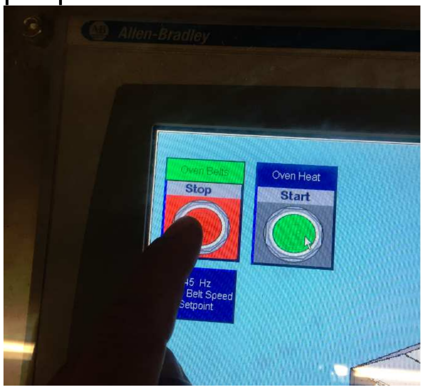

Figure 1.1

Turn the Heat on to light the oven pilots and main burners (Figure 1.2).

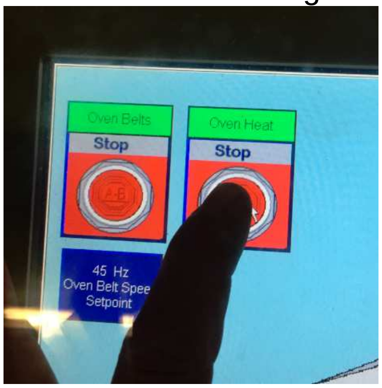

Figure 1.2

If the oven does not light but the belts turn, check to see that the gas valve is open (Figure 1.3). Go to the touchscreen. On the touchscreen, turn the heat on (Figure 1.2). The oven will light in 30 to 90 seconds, depending on the flame safety controller. The pilot flame will light when the igniter sparks (Figure 1.4). At the same time that the spark igniter goes on, the pilot gas solenoid valve opens (Figure 1.5).


Figure 1.3

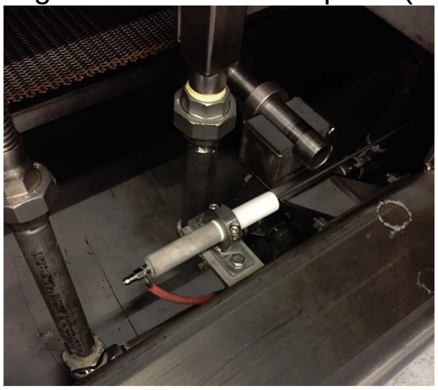
Figure 1.4

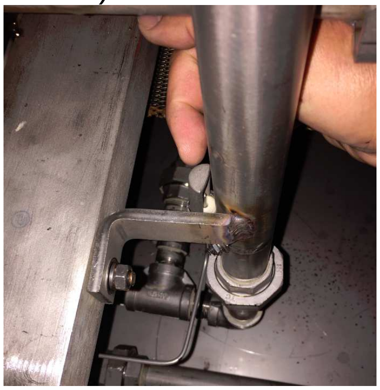
Figure 1.5

Once the pilot tree lights on the oven, the UV scanner will detect the flame of the pilot tree and send the signal to the flame controller (Figure 1.6). The flame controller will allow the main gas valves to open up. These will feed gas to the oven main burners on the top, middle and bottom decks of the oven.

Figure 1.6

Temperature Control and Oven Dwell

The temperature controllers will control the temperature in each deck using the setpoint established by the operator (Figure 1.7). The setpoint will be the main control for proper baking. Time will be the second main control for the baking. A signal will come from the PLC to an actuator motor which opens an air butterfly valve which controls the amount of gas into each deck of main burners (Figure 1.8). The actual temperature is critical and the operator must change the setpoint by using the keypad on the touchscreen (Figure 1.7). The speed of the bake is controlled by the speed of the belts in the oven (Figure 1.9). The touchpad will allow for the operator to establish the speed setpoint in minutes and seconds. This is important and is established by the operator. This will be the length of time the product is in the oven. Once the product enters the oven, the top belt transfers product to the middle belt. And the second turnaround has the product transferring from the middle deck to the bottom deck. The bottom deck will finish baking the product and the product will exit the oven. At the discharge of the oven there is a discharge conveyor which will transfer the baked product to the cooling conveyor.

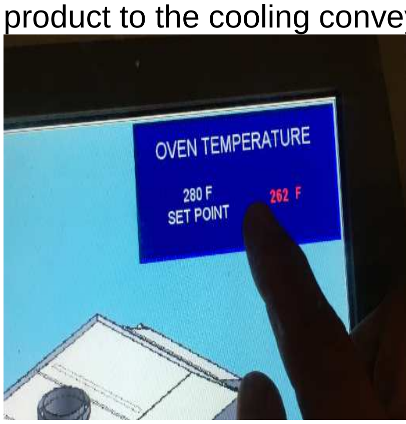

Figure 1.7

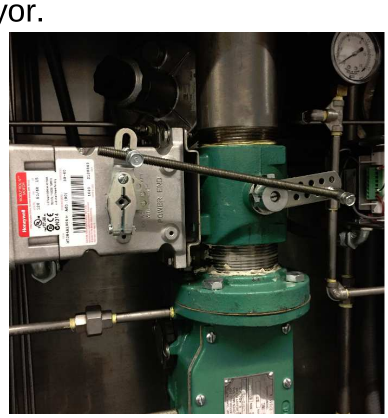

Figure 1.8

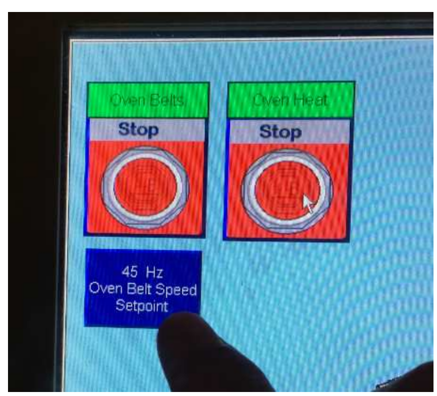

Figure 1.9

Product Quality and Appearance

The oven has three controls in total, one for each deck of the oven. Each one will have control of a single deck. We call the oven belts and/or burners, the top, middle and bottom (Figure 1.10). The butterfly valves control the amount of gas going to each deck of main burners. Each deck's product appearance must be inspected by the operator to see if the product and the baking of that product is acceptable on each deck. Each deck has different contact points to each side or face of the tortilla. One side is on the inside and the other on the outside of the tortilla. This is where the baking affects the look of the product. The top deck and bottom deck bake the same side of the product face. The middle deck touches the product face only one time and usually causes less burn marks or toast points. Temperature and time will affect the moisture and outside appearance of the product. The first turnaround or slide is a 'rake' slide (Figure 1.11 & 1.13). This rake slide may be enhanced with a piece of Teflon sheet or used press Teflon belting material which is clipped onto the rake slide frame (Figure 1.14). This will allow for easier transition from the top deck to the middle deck in the oven. These Teflon sheets also wear out and have to be replaced once the tortillas begins to stick. The rake slide can be adjusted to a position that prevents the folding of the tortillas. The second turnaround or second slide is nearer the press on the infeed side. This 2 piece slide is our patented 'double pull-out' slide (Figure 1.12). You can open the cover door to have access to the pullout slide and to inspect the slide position (Figure 1.15). The transition of the tortilla from the middle deck to the bottom deck can be seen through this area.

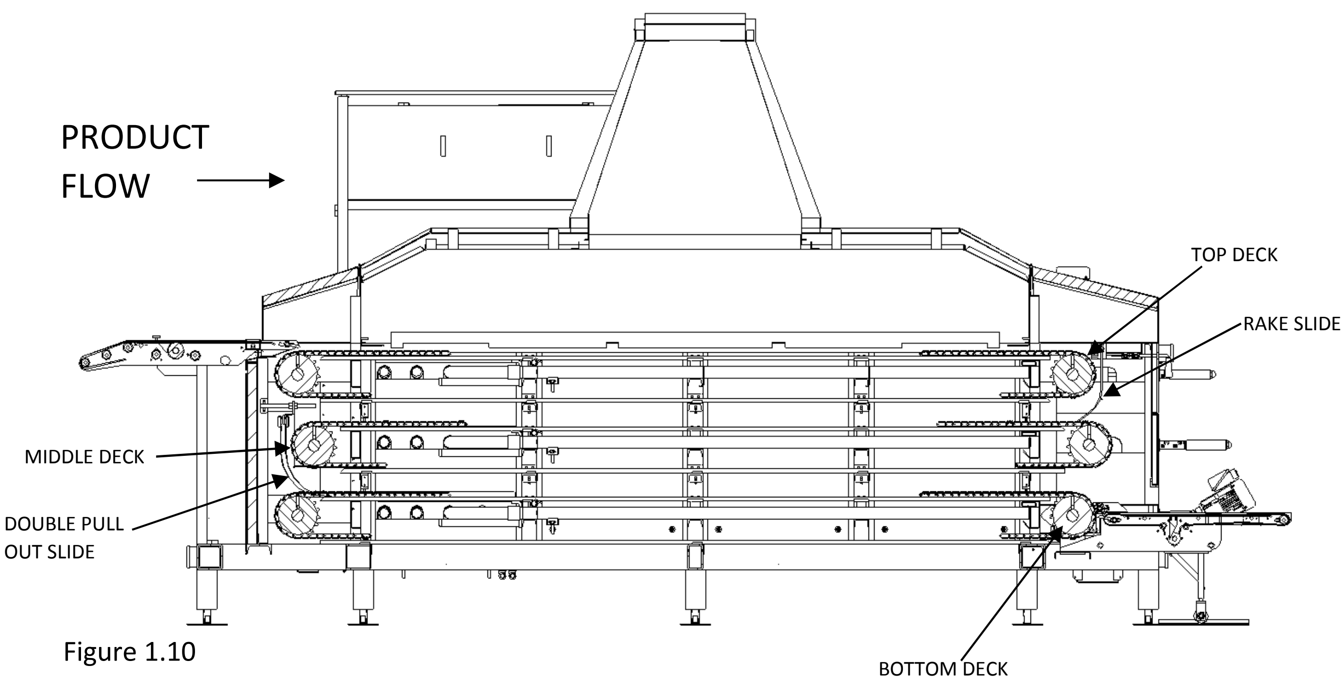

Figure 1.10

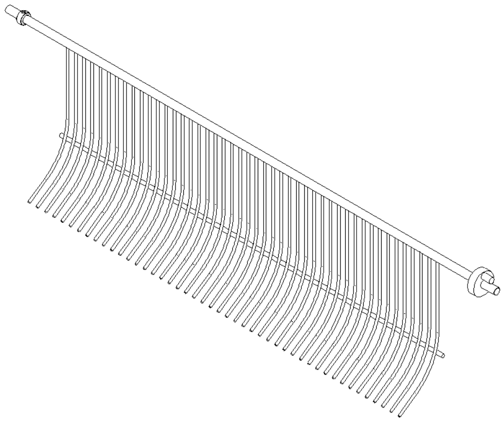

Figure 1.11

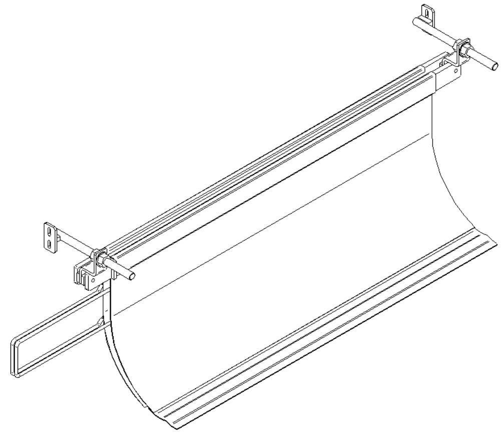

Figure 1.12

Figure 1.13

Figure 1.14

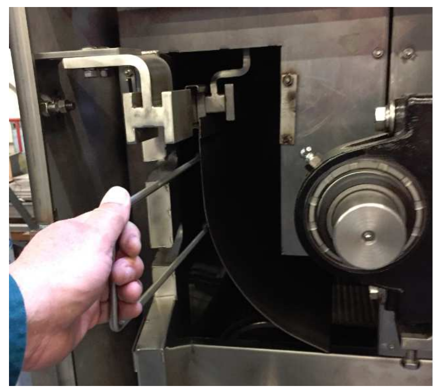

Figure 1.15

The final transfer point for the tortillas is at the discharge. The bottom belt to the cooling conveyor is also important. Tortillas can fold here as well. There is a flapper bar (Figure 1.16 & 1.17) that rotates and keeps the product from folding as it falls away from the oven. The tortillas will be full of steam and very soft and pliable.

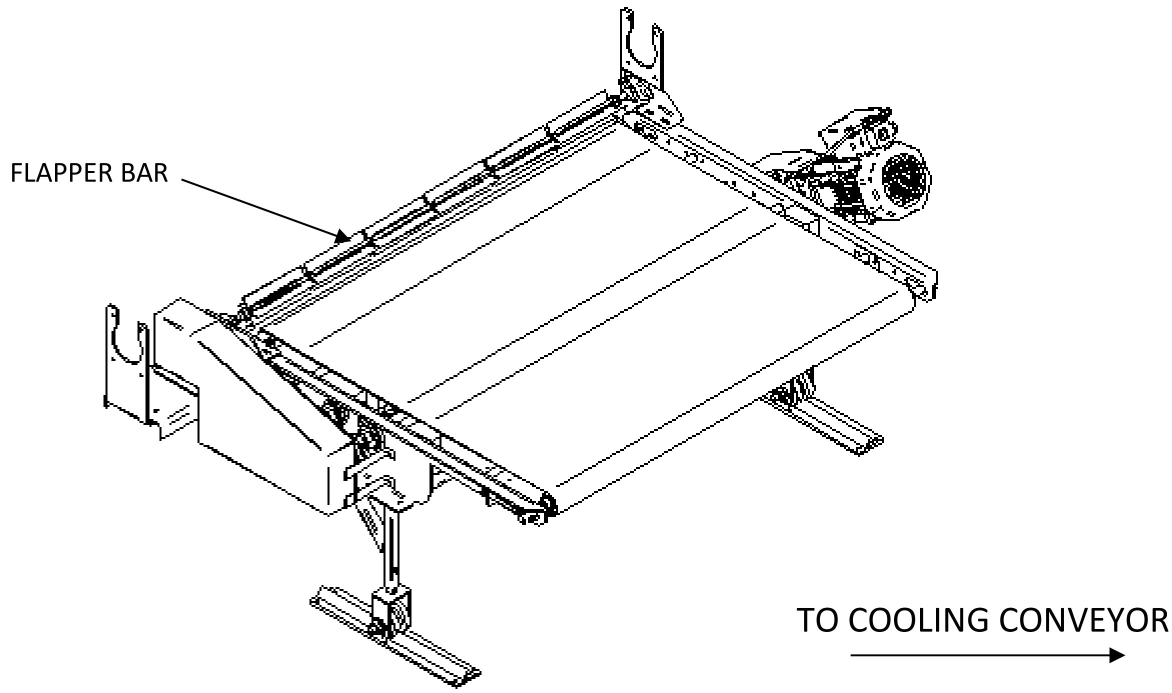

Figure 1.16

Figure 1.17

CHINS 190

PRE-MIX ODYSSEY OVEN OPERATION

PRE-OPERATION SAFETY CHECK

Check that all guards are in place and secured. Make sure that oven side panels and doors are in place and good condition and closed.

Look in both ends of the oven to see if anything had been left on any of the belts. Also, make sure that no one is working on or around the machine you are about to start.

STARTING AND OPERATION

1. Turn the ELECTRICAL MAIN DISCONNECT to "ON".

2. Turn MAIN POWER switch "ON".

3. Pull all emergency stops to "ON" position. Start oven drive from OIT (Operator's Interface Terminal). Pull main drive button to "ON" position.

4. Set the TEMPERATURE CONTROLLER to desired temperature.

5. Pull the COMBUSTION BLOWER switch to "ON" position.

6. Pull the HEAT START switch "ON". This starts a timed purge cycle. When completed, the spark igniter is energized and a 5-second trial for ignition begins. If the pilot flame is proven, the main burners will ignite within 10 seconds. The oven will rise to setpoint temperature.

NOTE: If the burners do not ignite or go out after lighting, push the RESET button for the FLAME SAFETY CONTROL and do again as directed in "6" above. If the burners do not stay lit after 3 tries, notify your supervisor or maintenance department before trying again.

DWELL TIME (BAKE BELT) ADJUSTMENT

Electrical panel: AC Variable-Frequency Drive feeds bake belt motor. Adjust speed pot at panel or digital OIT control. Slow down or speed up dwell time by controlling speed of bake belts.

MANUAL DWELL TIME (BAKE BELT) ADJUSTMENT

Oven main drive: Adjustable pulley motors have hand wheels to turn to speed up or slow down bake belts. Turning the handwheel **counter clock-wise** will *speed up* the drive and turning it **clockwise** will *slow down* the drive.

LONGITUDINAL BURNER ADJUSTMENT

1. With oven main burners LIT, open an oven door to visually inspect flame on **each deck's** left, left center, right center and right side main burners.

2. Adjust air/gas mixture at each deck's mixer to attain an efficient combustion. This is done with combustion air butterfly valve (that is mounted BEFORE mixer) FULLY open.

   Acotech Infrared burners- air/gas ratio has 19.3-19.6% percent oxygen
   Top deck burners-air/gas ratio is 18.9-19.1 percent oxygen; target 19.0
   Middle deck burners-air/gas ratio is 18.7-18.9 percent oxygen; target 18.8
   Bottom deck burners-air/gas ratio is 18.5-18.7 percent oxygen; target 18.6

3. During continuous production with the oven temperature stabilized, the TOP DECK baking belt's individual rows of product can be viewed by opening the front, side door and *viewing the product on the middle deck belt*. By comparing the toast points on the product from each row of TOP DECK, to the toast points on each row of product from the bottom deck, you will know which lane's butterfly valves to close down on TOP DECK.

4. During continuous production with the oven temperature stabilized, the MIDDLE and BOTTOM DECK'S product can be viewed by retrieving samples of each row as they exit the bottom deck oven discharge. By comparing row to row, you will know which lane's butterfly valves to close down. This product should be compared to the TOP DECK'S product.

5. Individual rows left, left center, right center and right side burners (Quad) or left and right burners (Double) should be adjusted during continuous production runs. The butterfly valves must be open as much as possible and close down the HOT LANES only.

6. Each burner is equipped with its own butterfly valve, (which are mounted AFTER the mixer) that controls the intensity and size of each flame. By loosening the thumbscrew, the butterfly valve can be closed or opened and locked into place when retightened. This will remove or add heat to their respective rows of product.

OVEN SHUT-OFF PROCEDURE

1. Allow all remaining product to finish its trip through the oven.

2. Push the HEAT START button to “OFF”.

3. Push the COMBUSTION BLOWER button to “OFF”.

4. Allow the oven belts to run until the oven belts have cooled to **250 degrees Fahrenheit or 121 degrees Celsius.** Typically, it is the top belt that is the last belt to cool down. Opening the oven doors will promote faster cooling. Slowing the belt speed promotes a quieter work environment.

5. Push the BELT MAIN DRIVE button to “OFF”.

6. Turn the ELECTRICAL MAIN DISCONNECT switch to “OFF” position.

7. Shut “OFF” gas supply at the HAND VALVE nearest the oven.